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Teaching materials and assessments will be uploaded to this QQ group only

Welcome to

Probability and Statistics

Fall, 2025

Textbook (try to get 1 of the 2)

- Bowerman, O'Connell, Murphree and Orris. ***Essentials of Business Statistics***

- Bruce L. Bowerman, Richard T. O' Connell , J.B. Orris, Emily S. Murphree
- Essentials of Business Statistics (Fifth Edition)
- ISBN 978-0-07-802053-7
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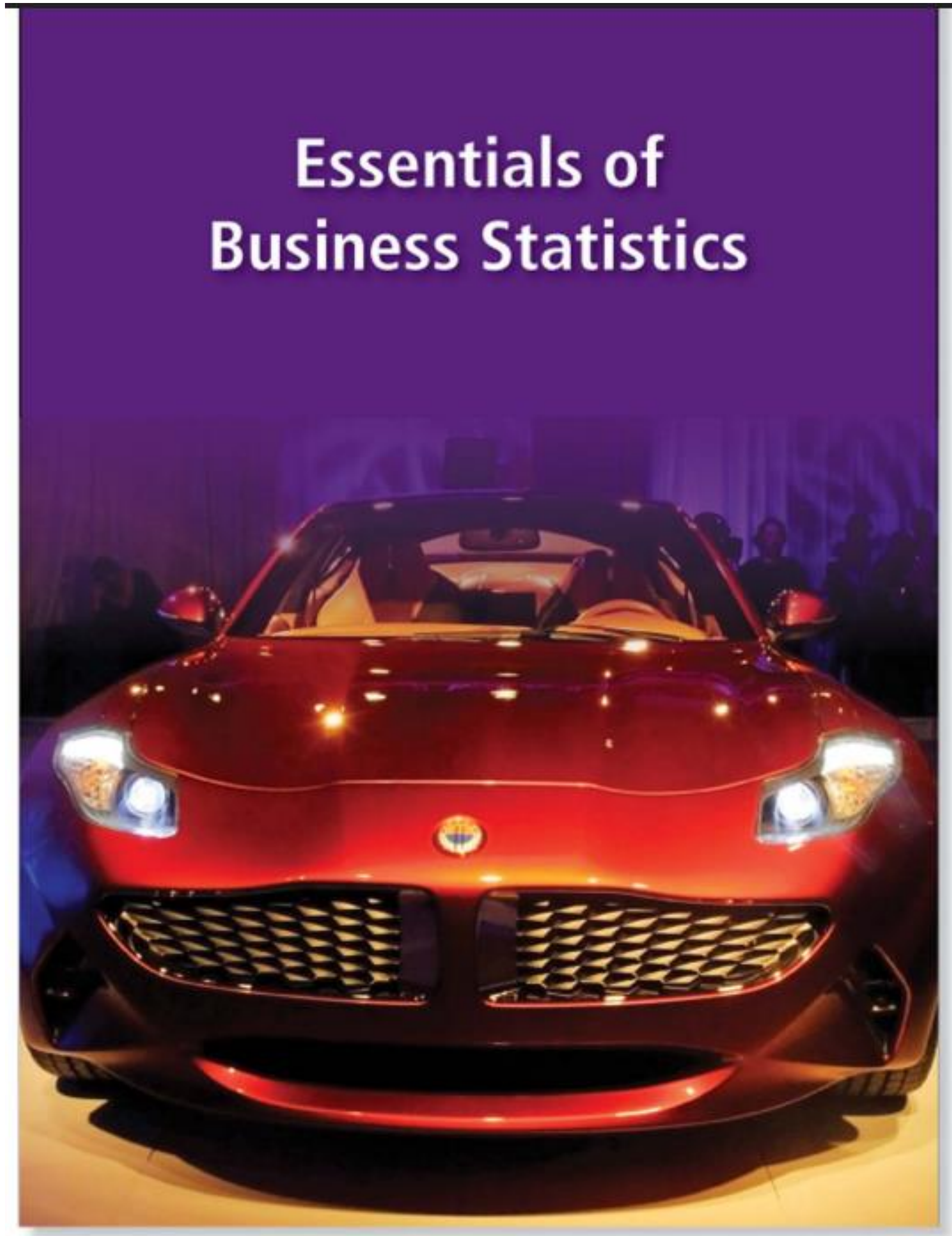
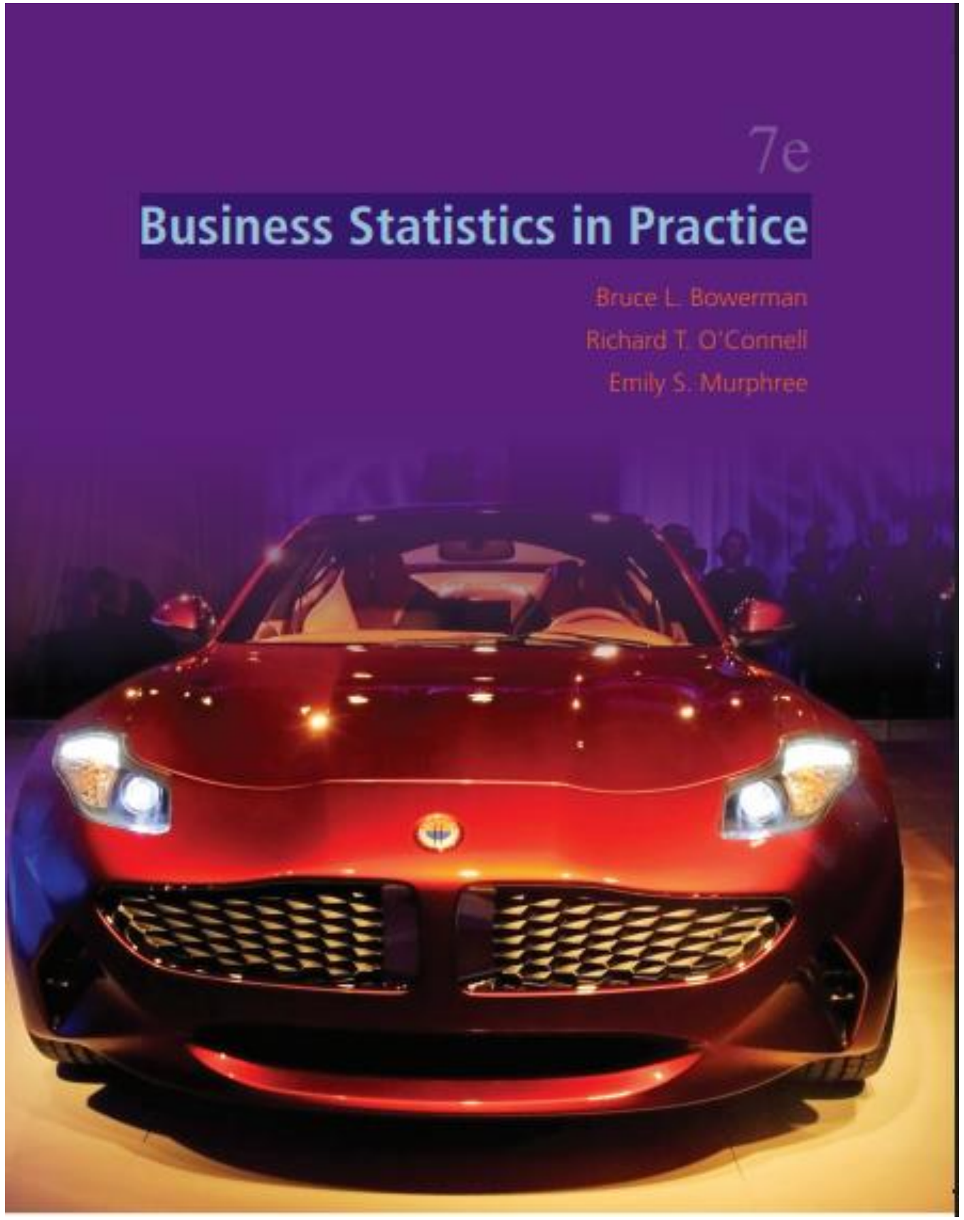


Textbook (try to get 1 of the 2)

- Bowerman, O'Connell, Murphree and Orris. ***Business Statistics in Practice***

- Bruce L. Bowerman, Richard T. O’ Connell , J.B. Orris, Emily S. Murphree
- Business Statistics in Practice (Seventh Edition)
- ISBN 978-0-07-352149-7
- Copyright © 2010 by McGraw-Hill Education (Asia) .



Reference

- D. Freedman, R. Pisani and R. Purves, Statistics, 3rd Ed., Norton, 1998.
- Chapter 1-18, 20-23, 26-28.
- W. Feller, Introduction to Probability Theory and Its Applications, Vol. 1, 3rd Ed., Wiley, 1968. page 1-258 J.E. Freund, Mathematical Statistics, 5th Ed., Prentice Hall, 1992, Chapter 1-9.
- Salsburg, The Lady Tasting Tea. How Statistics Revolutionized Science in the Twentieth Century, Freeman, 2001, Chapter 11-12
- Viktor Mayer-Schönberger and Kenneth Cukier, Big Data: A Revolution That Will Transform How We Live, Work, and Think, Eamon Dolan/Mariner Books, 2014.
- Charles Wheelan, Naked Statistics: Stripping the Dread from the Data, W. W. Norton & Company, 2014.
- Joel Best and Patrick Lawlor, Damned Lies and Statistics: Untangling Numbers from the Media, Politicians, and Activists, University of California Press; First Edition, 2012.
- Dana K. Keller, The Tao of Statistics: A Path to Understanding (With No Math), SAGE Publications, Inc, 2005.
- Gary Smith, Standard Deviations: Flawed Assumptions, Tortured Data, and Other Ways to Lie with Statistics, Overlook Hardcover, 2014.
- Matthew B. Robinson, Lies, Damned Lies, and Drug War Statistics, State University of New York Press, 2nd edition, 2014.
- Introduction to Mathematical Statistics. ROBERT V.HOGG， ALLEN T. CRAIG
- Essentials of Business Statistics. Bowerman, O'Connell, Murphree and Orris. 2015. McGraw-Hill/Irwin. ISBN10: 0078020530.
- Business Statistics in Practice. Bowerman, O'Connell, Murphree and Orris. McGraw-Hill/Irwin. ISBN 978-0-07-352149-7.

Reading List

- Darrell Huff (1991) *How to Lie with Statistics* New Ed edition, ISBN 0-14-013629-0
- Rao C R. (1997) *Statistics and truth: putting chance to work* World Scientific.
- Salsburg, D. (2002) *The Lady Tasting Tea: How Statistics Revolutionized Science in the Twentieth Century*, W.H. Freeman / Owl Book. ISBN 0-8050-7134-2
- Reinhart A. (2015) *Statistics done wrong: The woefully complete guide* No Starch Press.
- Moore S. and Notz W. (2017) Statistics: Concepts and Controversies 9th Edition.

Course Description

This course is designed to provide undergraduate students with a comprehensive understanding of statistical concepts and their applications. This course aims to develop students' quantitative reasoning skills and equip them with the tools necessary to analyze and interpret data for decision-making.

Course Objectives

- Understand the fundamental concepts of descriptive and inferential statistics.
- Learn various techniques for data collection, organization, and presentation.
- Develop the ability to analyze and interpret data using appropriate statistical methods.
- Apply statistical tools to solve real-world problems.
- Enhance critical thinking skills by evaluating the validity and reliability of statistical results.
- Basic probability and statistical concepts and methods.
- Emphases on what, how, when and why certain statistical methods can and cannot be applied.
- Solving simple real-life problems by statistics.

Course Topics

1. Introduction to Statistics: Importance and role in decision-making, ethical considerations.

2. Descriptive Statistics: Measures of central tendency, variability, and graphical representation of data.

3. Probability: Basic concepts, probability distributions, and their applications.

4. Sampling and Estimation: Sampling techniques, confidence intervals, and sample size determination.

5. Correlation and **Regression** Analysis: Relationship between variables, linear regression, and interpretation of regression models.

Assessments

Assignments
(30%)

Quizzes & Attendance
(10%)

Project
(20%)

Final examination
(40%)

Assignment (30%)

- 5 assignments in total, assignment questions will be provided in weeks 2, 4, 6, 8, 10.
- Please complete and hand in your assignments in weeks 3, 5, 7, 9, 11.
- You are encouraged to discuss homework problems with others or ask the lecturer (me), but the must write your solution **independently**.
- **Duplicated** homework or solution with no supporting work receives **no credit**.
- The assignments will be Written Answers: - Use the correct method to answer the question 80%; Get the correct numerical value(s) 20%.

Quizzes & Attendance (10%)

- 2 quizzes, during the teaching session (in the class)
- Each in a week that is randomly selected.
- Quiz is in the class.
- 20 minutes each quiz
- Complete and hand in your quiz solutions immediately after the quiz.
- Work **independently** – no discussion.
- The assignments will be Written Answers: - Use the correct method to answer the question 80%; Get the correct numerical value(s) 20%.

Project (20%) – one (1) report

- Assignment will be provided to you some time in week 6
- Hand your report (hard copy, hand-writing or electronic writing) 2 weeks after.
- Everyone submits one report.
- 300 ~ 500 words (not including figures, tables, equations)
- Report should have following structure (title, your name and your student ID, abstract, introduction (background), analysis, conclusion, acknowledges (if you have).

Project – marking criteria

- The structure of the report
- Presentation of aims, results and conclusions
- Writing English
- Handwriting must be clear
- Electronic writing must be print it out, as only hard copy is accepted.
- Question (can answer the question correctly).

Final exam (40%)

- Final exam will be Written Answers:

  - Use the correct method to answer the question 80%;

  - Get the correct numerical value(s) 20%.

- Final exam will be close-book.
- Final exam (day, time, location) will be announced by the university.

Use of Artificial Intelligence:

Please note:

1. Use of generative artificial intelligence (AI) is **NOT** permitted in any assessments of this subject.

2. Working with anyone else or using AI technology will get an unfair advantage over other students in the assessments and is therefore **NOT** tolerated.

3. Use of AI tools may be detected.

Rules in classes

- The university's order is that you need to sign in your attendance every session. However, your attendance does not affect your assignment marking.

- **No** video, **No** voice recording (we respect the privacy for all) – Offended will be warning and even be asked to leave the class.

- If you come late, entering in the classroom from the rear door quietly.

Rules in classes

- Be quiet in the class, **no** speaking, **no** talking, **no** answering phone.
- Ask questions after the class, NOT in the class.
- Every class, I will leave ~10 minutes for open discussion and for Q&A. Yes, you can discuss / talk at this time.

What is Statistics all about?

- **Statistics** is "learning from data" – the science of designing studies and analysing their results
- Statistics is concerned with
  - ➢ the collection of data
  - ➢ their description or summarize.
  - ➢ their analysis, which often leads to the drawing of conclusions (interpretation)

Data are numerical facts and figures from which conclusions can be drawn. Such conclusions are important to the decision-making processes of many professions and organizations.

Examples:

- How many sisters / brothers do you have?
- How often do you exercise in one week? ----Never (0); rarely (2); often (4); always (7)
- How are you ? ---- Terrible (0), not bad (2); I am fine (5).

Why Study Statistics?

- To meet Graduation requirement
- Generally, make "decision"
- Statistical techniques are used to make decisions that affect our daily lives (without knowing it)
  - ➢ Choose the university
  - ➢ To buy a car or a house
  - ➢ Choose common stock from the stock markets
- No matter what your career, you will make professional decisions that involve data.
- Joy of Statistics https://www.bilibili.com/video/BV1TJ411y7Mp?from=search&seid=1904466107273049131

Everyone has **data** that they need to analyze and we now live in an age of data!

Statistics plays a major role in psychology, social science, biology, chemistry, engineering, physics, economics / finance, medical science, computer science, aviation, education, sports, robotics, AI, etc.

Example 1: A survey is conducted among 50 people, asking them: “Which one of the following is your favorite, Orange juice(O), Apple juice(A), Coca Cola(C), Pepsi(P) or Coconut Juice (L) ? ”

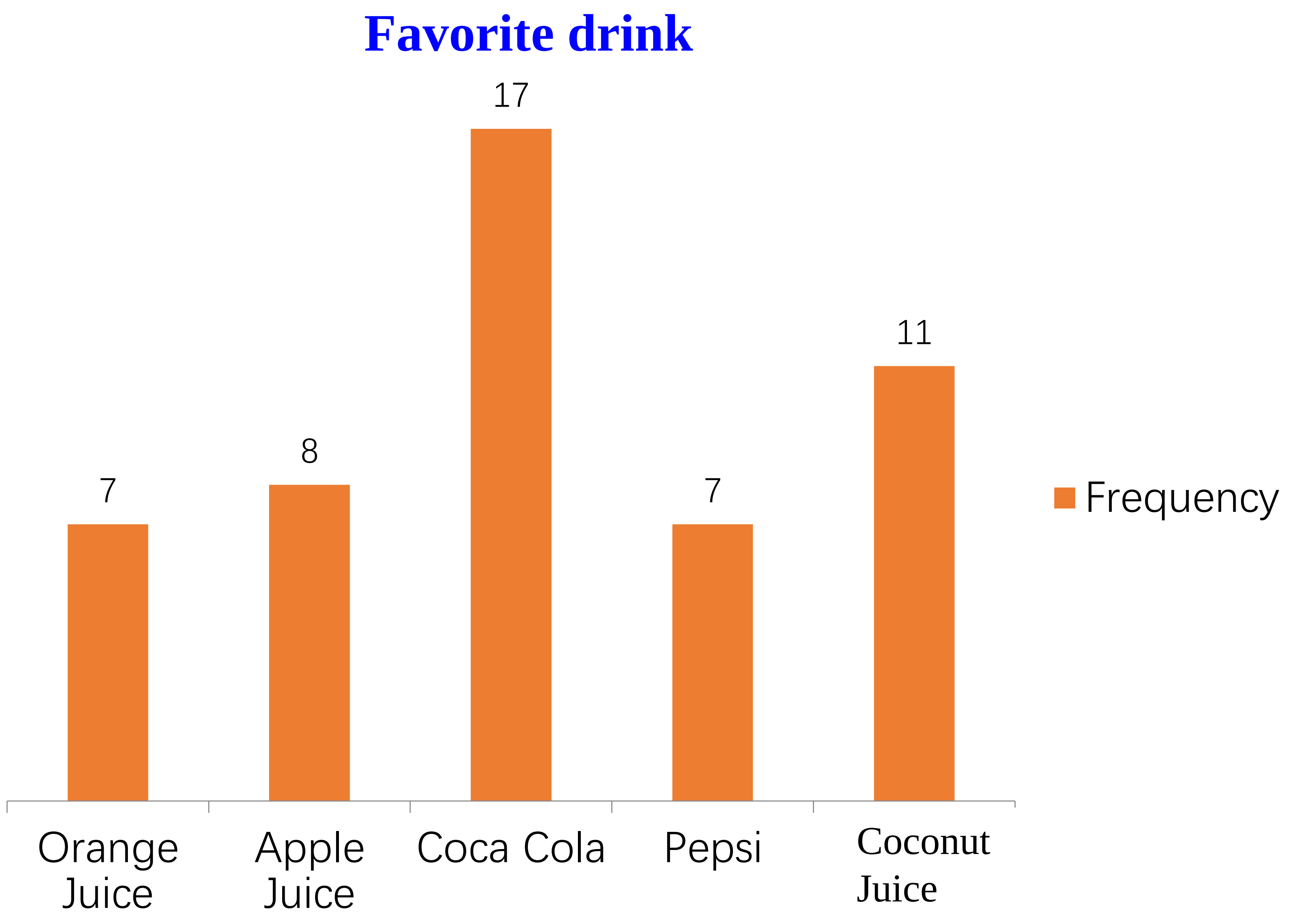
Data collected are

O, A, C, A, C, P, C, A, C, C, C, A, L, O, C, P, A, O, L, P, L, O, C, L, P, O, O, C, A, L, C, L, L, C, C, L, C, L, P, O, C, A, P, A, C, C, L, L, C, P

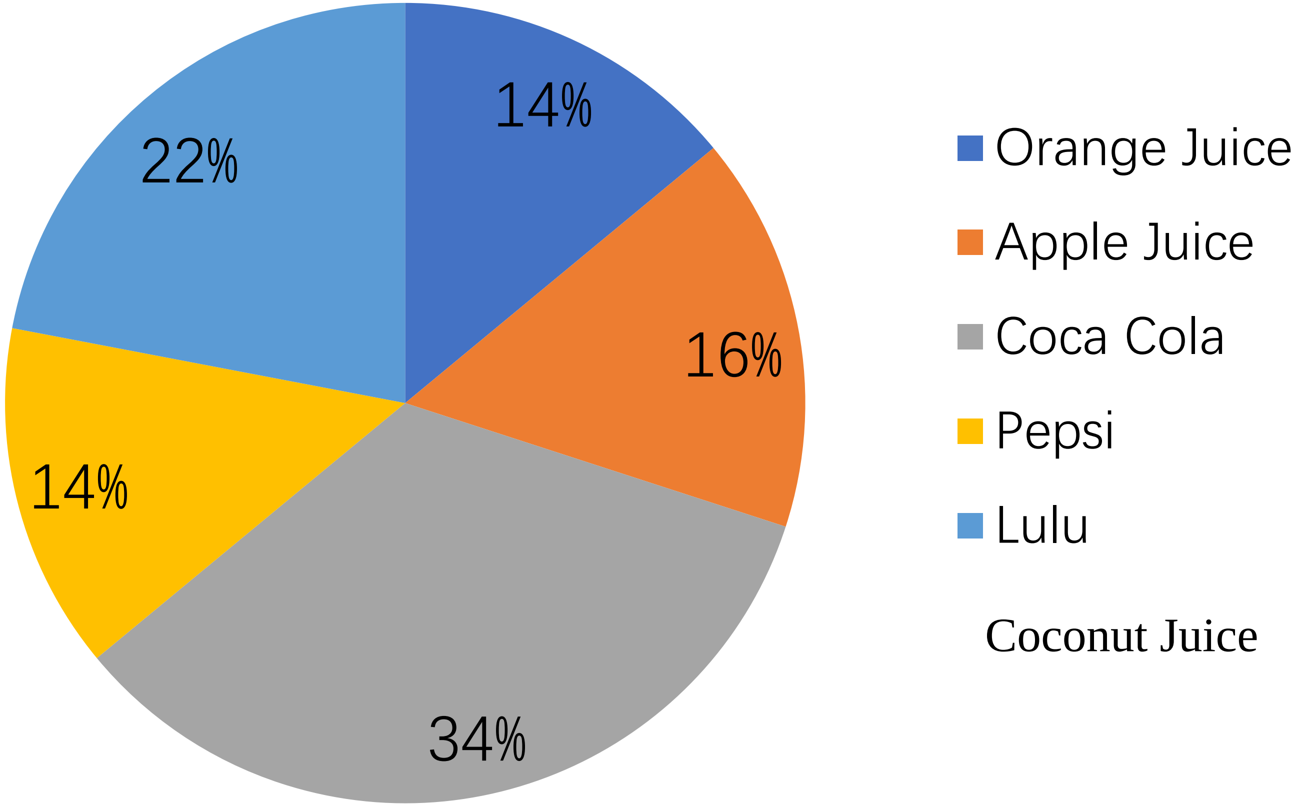
Questions: Which one is the most popular?
Which one is the least popular?

Answer:

Table: Frequency and percentage of favorite drink		
Drink	Frequency	Percentage
Orange Juice	7	14
Apple Juice	8	16
Coca Cola	17	34
Pepsi	7	14
Coconut Juice	11	22
Total	50	100


Favorite drink
17
11
8
7
7
Frequency
Orange Juice
Apple Juice
Coca Cola
Pepsi
Coconut Juice

Favorite drink



Coconut Juice

Example 2: A poll was held asking a selection of high school students which major (Engineering or Social Science) they would choose for their university study.

Out of 400 respondents, 63% said they would choose Engineering ahead of Social Science.

Some important questions where statistics can help are:

- How accurate is this estimate of the proportion of people voting for Engineering ahead of the Social Science?
- Is this sample, of just 400 students, sufficient to predict that most students would choose Engineering for their university major ?

More Examples:

- Studying whether a new treatment works.
- Investigation climate change effects on species distributions
- Your chance of winning the lottery. Is it worth playing?
- Is there gender bias in promotion?
- Checking if Shenzhen getting less rainfall than it used to.
- Understanding the effects of a new pest on wildlife.

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Think

- What will we learn in this course?
- How to use them in real life data analysis?
- What else can we do? (innovation)

Statistics in the real-world

Ever wonder what's really going on behind the news headlines, advertisements, or political polls? Statistical reasoning provides a powerful lens to critically evaluate information and gain a deeper understanding of the world around you.

In fact, it has been argued that understanding of statistical is essential to truly understanding the world around you.

Beyond everyday understanding, statistics is of fundamental importance across a vast range of disciplines. Consequently, statistical skills are highly valued, and there's a significant shortage of statisticians in the job market – great news if you decide to major in statistics!

Furthermore, the recent explosion in artificial intelligence (AI) and machine learning has placed statistics at center stage. The very algorithms powering AI are built upon core statistical principles, making this field an exciting frontier for those with statistical expertise.

We’ll start by building a strong foundation with key statistical concepts and terminology (such as Data, Population, Sample, Census, Description, Inference, Random sample). These fundamental ideas will reappear throughout this course and are central to the entire field of statistics.

So, let’s begin our journey!

Thank you!